

Program–8: Create Kubernetes Pods using YAML Descriptors

Program structure:

Program-8/

- ├── server.js
- ├── nodejs-configmap.yaml
- └── nodejs-pod.yaml

Step 1: Write Node.js app

nano server.js

```
GNU nano 7.2 server.js
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Create ConfigMap YAML

nano nodejs-configmap.yaml

```
GNU nano 7.2 nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create Pod YAML

nano nodejs-pod.yaml

```
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Apply the YAML files

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml

Step 5: Check pod status

kubectl get pods

Step 6: Access the Node.js app

kubectl port-forward pod/nodejs-pod 3000:3000

```
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ minikube start
👉 minikube v1.37.0 on Ubuntu 24.04
🔗 Using the docker driver based on existing profile
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: metrics-server, storage-provisioner, dashboard, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ nano server.js
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ nano nodejs-configmap.yaml
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ nano nodejs-pod.yaml
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config configured
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nodejs-pod    1/1     Running   0           28s
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

Step 7: Check output in browser

http://localhost:3000

